

Chapter 8:

Integration Techniques and Improper Integrals

Section 8.5: Partial Fractions

In this section we evaluate integrals of the form $\int \frac{P(x)}{Q(x)}\,dx$ where P and Q are polynomials.

EXAMPLE

$$\int \frac{4x^2 + 13x - 9}{x^3 + 2x^2 - 3x}\,dx$$

$$\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1}\,dx$$

Using the techniques of this section we will be able to evaluate other forms of integrals for example

$$\int \frac{dx}{x^2 + x\sqrt{x}} \quad \text{and} \quad \int \frac{\sec^2 x}{\tan^2 x + 5\tan x + 6}\,dx$$

The Method of Partial Fractions

Definition: The rational functions of the form $\dfrac{A}{(ax+b)^r}$ *or* $\dfrac{Ax+B}{(ax^2+bx+c)^r}$, where A, B, a, b, c are real constants and with $b^2-4ac<0$ and r is a positive integer, are called **partial fractions**.

The idea of this method is to express a rational function as a sum of partial fractions.

EXAMPLE

If you know that $\dfrac{4x^2+13x-9}{x^3+2x^2-3x}=\dfrac{3}{x}-\dfrac{1}{x+3}+\dfrac{2}{x-1}$ then evaluate the integral $\displaystyle\int \frac{4x^2+13x-9}{x^3+2x^2-3x}\,dx.$

The Method of Partial Fractions

Procedures of Partial Fractions Method for the Rational Function of the form $\frac{P(x)}{Q(x)}$:

Step 1: If $\deg P \geq \deg Q$, **do the long division**, otherwise go to the next step.

Assume that $\frac{P(x)}{Q(x)} = S(x) + \boxed{\frac{R(x)}{Q(x)}}$. ← We focus on this term.

Step 2: If $Q'(x) = R(x)$, then the result of integration $\int \frac{R(x)}{Q(x)} dx = \ln|Q(x)| + C$. If $Q'(x) \neq R(x)$ we go to the next step.

Step 3: Factor $Q(x)$ as a product of linear factors that are possibly repeated $(a_1x + b_1)^r$, irreducible factors that possibly repeated $(a_2x^2 + b_2x + c)^m$ with $b_2^2 - 4a_2c < 0$.

EXAMPLE

(1) $Q(x) = x^2 - 1 = (x - 1)(x + 1)$.

(2) $Q(x) = x^3 + x = x(x^2 + 1)$.

The Method of Partial Fractions

Procedures of Partial Fractions Method for the Rational Function of the form $\frac{P(x)}{Q(x)}$:

Step 4: Express $\frac{R(x)}{Q(x)}$ as a sum of partial fractions as follow:

- For each factor $(a_1x + b_1)^r$, the corresponding partial fraction decomposition is

$$\frac{A_1}{a_1x + b_1} + \frac{A_2}{(a_1x + b_1)^2} + \ \ldots + \frac{A_r}{(a_1x + b_1)^r}$$

- For each factor $(a_2x^2 + b_2x + c)^m$, the corresponding partial fraction decomposition is

$$\frac{A_1x + B_1}{a_2x^2 + b_2x + c} + \frac{A_2x + B_2}{(a_2x^2 + b_2x + c)^2} + \ \ldots + \frac{A_mx + B_m}{(a_2x^2 + b_2x + c)^m}$$

The Method of Partial Fractions

Procedures of Partial Fractions Method for the Rational Function of the form $\frac{P(x)}{Q(x)}$:

Step 5: Find the unknown **coefficients** $A_1, A_2, \dots, B_1, B_2, \dots$.

- The general way in finding the coefficients is to multiply the both sides of the equation

$$\frac{R(x)}{Q(x)} = \ sum\ of\ partial\ fractions$$

by the denominator $Q(x)$ and equating the corresponding coefficients of x^i. Then we solve the resulting system.

- In some cases, substitution for x will give instant values of some of the coefficients.

- If the partial fractions are for **distinct linear factors**, we can use "**Cover-up**" method (or Heaviside method) to find the corresponding coefficients.

All the above methods will be explained in the following examples.

EXAMPLE

Find $\int \frac{4x^2 + 13x - 9}{x^3 + 2x^2 - 3x}\, dx.$

SOLUTION

$$I = 3\ln|x| + 2\ln|x-1| - \ln|x+3| + C$$

EXAMPLE

Find $\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1}\, dx.$

The long division leads to $\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = x + 1 + \frac{4x}{x^3 - x^2 - x + 1}.$

SOLUTION

$$I = \frac{x^2}{2} + x - \frac{2}{x-1} + \ln|x-1| - \ln|x+1| + C$$

Remark: The following integrals may be needed when applying the partial fractions decomposition

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} arctan\left(\frac{x}{a}\right) + C\,.$$

$$\int \frac{dx}{ax + b} = \frac{1}{a} \ln|ax + b| + C.$$

$$\int \frac{f'(x)dx}{f(x)} = \ln|f(x)| + C.$$

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln\left|\frac{x - a}{x + a}\right| + C.$$

EXAMPLE

Find $\int \frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)}\, dx.$

SOLUTION

$$I = \frac{3}{2}\ln(x^2+4) - \frac{5}{2}\ln|2x-1| + \frac{1}{2}\arctan\left(\frac{x}{2}\right) + C$$

EXAMPLE

Find $\int \dfrac{3x^2 - x - 21}{x^3 - \frac{1}{2}x^2 - 21x + 5}\, dx.$

SOLUTION

EXAMPLE

Express the function as a sum of partial fractions. Do not evaluate the coefficients

$$\frac{x^2 + 2x - 1}{(x+3)^3(x^2+2x+2)(x^2+2)^2} =$$

EXAMPLE Do the Substitution First

Find $\int \frac{\sec^2 x}{\tan^2 x + 5\tan x + 6}\,dx.$

SOLUTION

$$I = \ln\left|\frac{2+\tan x}{3+\tan x}\right| + C$$

EXAMPLE

Rationalizing Substitution

Find $\int \frac{dx}{x^2 + x\sqrt{x}}$.

SOLUTION

$$I = -\frac{2}{\sqrt{x}} + 2\ln(1+\sqrt{x}) - \ln|x| + C$$

Solve exercises of section 8.5.

Start with the suggested problems mentioned in the syllabus.